BISHAL PAL

Software Engineer — Full Stack | Backend Systems | Distributed Applications

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EDUCATION

Master of Computer Applications (MCA), Adamas University, Kolkata *2024 – 2026*
CGPA: 7.5

B.Sc. in Computer Science (Honours), Bankura University *2021 – 2024*
CGPA: 7.6

PROFESSIONAL EXPERIENCE

Software Developer Intern (Blockchain) — Block Se Block *Jul 2025 – Sep 2025 | Remote*

- Engineered and deployed 5+ Solidity smart contracts (ERC-20, NFT minting) to Ethereum testnet, reducing gas costs and deployment time by 40% through bytecode and gas-usage optimization.
- Built a full-stack DApp with MetaMask authentication and ethers.js transaction handling for on-chain interactions.
- Implemented on-chain event listeners for real-time state updates and achieved 90%+ test coverage using Hardhat and Chai unit testing frameworks.

Software Developer Intern — Bluestocks *May 2025 – Jul 2025 | Remote*

- Shipped MERN stack features for a stock market simulation platform supporting 1,000+ active users, improving Lighthouse performance score from 62 to 89 (44% increase).
- Optimized MongoDB aggregation pipelines, cutting average API response time by 35% under production load.
- Refactored core React/TypeScript components, reducing bundle size and improving render performance across 10+ pages.
- Collaborated in Agile sprints with daily standups and bi-weekly code reviews across a 6-engineer team.

PROJECTS

StudyPilot AI — AI Study Coaching Platform — *Next.js, TypeScript, MongoDB, LangChain, OpenAI/Gemini API*

- Built a full-stack SaaS platform with adaptive quizzing, an SM-2 spaced-repetition algorithm, and AI-generated learning roadmaps powered by multi-LLM (OpenAI/Gemini) backends.
- Designed a RAG pipeline using LangChain to improve contextual Q&A accuracy for personalized study assistance.
- Implemented a live progress dashboard with real-time analytics to track user learning outcomes.
- Structured MongoDB schemas and API routes in Next.js/TypeScript to support scalable multi-user session handling.

AI Transaction Analytics Pipeline — Async Data Processing System — *FastAPI, Celery, PostgreSQL, Redis, Docker*

- Architected an async ingestion pipeline processing CSV transaction uploads via Celery workers, using Redis as a message broker for high-throughput batch jobs.
- Built an LLM-based transaction classifier with rule-based fallback and anomaly detection to flag suspicious patterns and reduce false-positive rates.
- Designed the PostgreSQL schema and query patterns to support fast aggregation over large transaction datasets.
- Deployed the system on Docker with SSL-secured, JWT-authenticated endpoints for production-grade security.

Solana DEX — On-Chain Order Book Engine — *Rust, Anchor, Next.js, WebSockets*

- Developed a native Solana program using PDAs for order-book state management and CPIs for token transfers.
- Implemented real-time limit-order matching logic on-chain for a decentralized order book engine.
- Built a Next.js frontend with Phantom wallet integration for seamless user authentication.
- Integrated live WebSocket account subscriptions and deployed the full DApp to Devnet for testing.

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, Python, Java, Go, Rust, C, C++

Frontend: React, Next.js, Redux, Tailwind CSS, Framer Motion, Responsive/Accessible UI

Backend: Node.js, Express, NestJS, Spring Boot, FastAPI, Flask, Django, REST APIs, Microservices, GraphQL-ready

Databases: PostgreSQL, MongoDB, Redis, DuckDB, Prisma ORM, Drizzle ORM, Query Optimization

Cloud & DevOps: Docker, Kubernetes, GitHub Actions (CI/CD), Nginx, Linux, Celery, Horizontal Pod Autoscaling, System Design

Distributed Systems: Blockchain (Ethereum, Solidity, Solana/Rust/Anchor), Event-Driven Architecture, Message Queues, WebSockets

Testing & Tools: JUnit, Mockito, Hardhat, Chai, Git/GitHub, Postman, Agile/Scrum, Code Review

ACHIEVEMENTS & ACTIVITIES

- IBM SkillsBuild AI Builders Challenge (2025): built a Python autonomous driving agent for the TORCS simulator.
- Active Solana Colosseum hackathon participant; published open-source Solana programs on GitHub.